Agentic AI Roadmap (Beginner to Advanced)

Phase 1: Python Foundations (Weeks 1–3)

Module 1: Python Basics

- Introduction to Python
- Variables and Data Types
- Input/Output
- Operators
- Type Casting
- Comments

Module 2: Control Structures

- If, Else, Elif
- Loops (For, While)
- Break, Continue, Pass
- Pattern Programs

Module 3: Functions

- Function Creation
- Parameters and Arguments
- Return Statements
- Lambda Functions
- Recursion

Module 4: Data Structures

- Lists
- Tuples
- Sets
- Dictionaries
- String Manipulation

Practice Projects

- Calculator
- Student Management System
- Password Generator
- Quiz Application

Phase 2: Advanced Python (Weeks 4–5)

Object-Oriented Programming

- Classes and Objects
- Constructors
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction

Exception Handling

- Try-Except
- Finally
- Custom Exceptions

File Handling

- TXT Files
- CSV Files
- JSON Files

Modules & Packages

- Built-in Modules
- Creating Custom Modules
- Virtual Environments

Practice Projects

- Library Management System
- Bank Management System
- Inventory System

Phase 3: Data Analysis Foundations (Week 6)

NumPy

- Arrays
- Indexing
- Broadcasting
- Mathematical Operations

Pandas

- DataFrames
- Cleaning Data
- Missing Values
- Filtering

- Grouping

Data Visualization

- Matplotlib
- Seaborn
- Plot Types

Project

- Sales Analysis Dashboard

Phase 4: Databases (Week 7)

SQL Fundamentals

- Create Database
- CRUD Operations
- Joins
- Aggregations

Python + MySQL

- Database Connectivity
- CRUD using Python

Project

- Student Record Management System

Phase 5: AI Fundamentals (Weeks 8–9)

Introduction to AI

- AI vs ML vs DL
- Types of AI
- Generative AI

Machine Learning Basics

- Supervised Learning
- Unsupervised Learning
- Classification
- Regression

Scikit-Learn

- Data Splitting
- Model Training
- Evaluation

Projects

- House Price Prediction
- Diabetes Prediction
- Spam Detection

Phase 6: Generative AI (Weeks 10–11)

Introduction to LLMs

- What are LLMs?
- Transformer Architecture
- Tokens
- Embeddings
- Context Windows

Prompt Engineering

OpenAI APIs

- API Integration
- Chat Completions
- Function Calling

Projects

- AI Chatbot
- Resume Analyzer
- PDF Question Answering

Phase 7: Retrieval-Augmented Generation (RAG) (Week 12)

RAG Architecture

$$f(x) = \text{LLM}(\text{Prompt} + \text{Retrieved Context})$$

Topics

- Vector Databases
- Embeddings
- Semantic Search
- Chunking
- Retrieval Pipelines

Tools

- LangChain
- FAISS

- ChromaDB

Projects

- College Knowledge Bot
- Company Policy Assistant
- Research Paper Assistant

Phase 8: Agentic AI Fundamentals (Week 13)

What is an AI Agent?

- Perception
- Reasoning
- Planning
- Action
- Memory

Concepts

- Agent Loop
- Tool Calling
- Memory Systems
- Multi-Step Reasoning

Frameworks

- LangChain Agents
- LangGraph
- CrewAI

Phase 9: Building Agents (Week 14)

Single Agent Systems

Projects

- Email Assistant
- Research Agent
- Travel Planner
- Meeting Summarizer

Tool Integration

- Calculator
- Web Search
- Database
- APIs

Phase 10: Multi-Agent Systems (Week 15)

CrewAI

Agent Types

1. Manager Agent
2. Research Agent
3. Writer Agent
4. Reviewer Agent

Workflow

Research → Analysis → Report Generation

Projects

- Cybersecurity Analyst Team
- Business Analyst Team
- Content Generation Team

Phase 11: Autonomous Agents (Week 16)

Advanced Agentic AI

Topics

- Planning Agents
- Reflection Agents
- Memory Agents
- Autonomous Workflows
- Human-in-the-Loop Systems

Frameworks

- CrewAI
- LangGraph
- AutoGen
- OpenAI Agents SDK

Projects

- Autonomous Research Assistant
- Cybersecurity Investigation Agent
- AI Recruitment Agent
- AI Customer Support Agent